How Blue is California?

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Research Question, Hypothesis, Variables, and Data

Research Question and Hypothesis

Research Question: Which demographic and socioeconomic factors are most strongly associated with Republican vote share across California counties?

Hypothesis: While demographic factors such as age, race, and education level will significantly predict Republican vote share across California counties, industry of employment (especially agriculture) will exert an outsized effect.

Independent and Dependent Variables

Independent variables:

Demographic

- Race
- Median age

Socioeconomic

- Education: % Bachelor’s degree+
- Income: median household income
- Unemployment rate
- Employment by industry

Dependent variable:

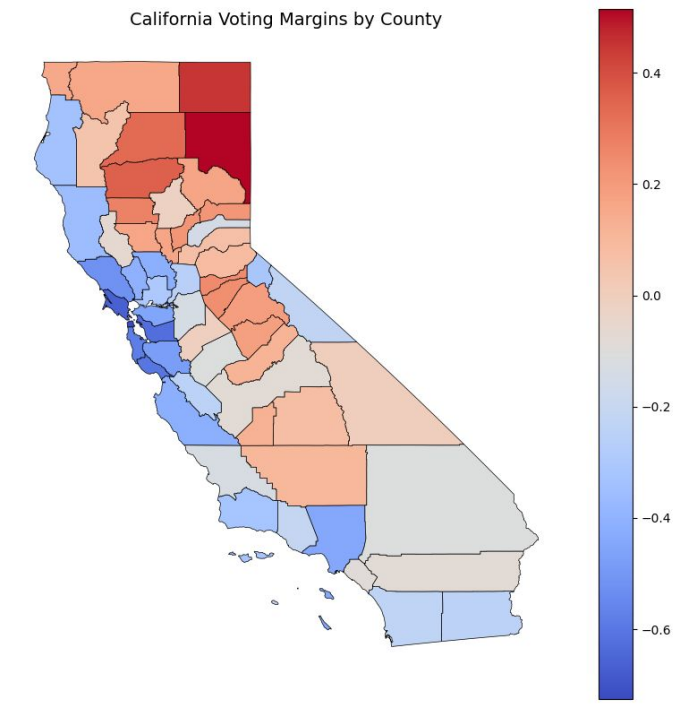
County level Republican vote share (2024 presidential election)

Datasets, Sources, and Cleaning Process

Data Source	Acquired From	Cleaning Process
Multiple American Community Surveys (ACSDP02, ACSDP03, ACSDP05) (59 x 24)	United States Census Bureau	Selected estimate and % columns, cleaned county names, droped metrics, made counties OU, merged
MIT Presidential Election Results by County (406 x 7)	MIT	Filtered to California presidential data, standardized party labels and formatting, removed unnecessary columns, created key variables (vote_share, dem_share, rep_share, margin, winner)
California Proposition Voting Results 2024 (58 x 21)	California Office of the Secretary of State	Removed county voting totals row, converted Yes/No votes to percentages, created outcome columns

Analysis (So Far)

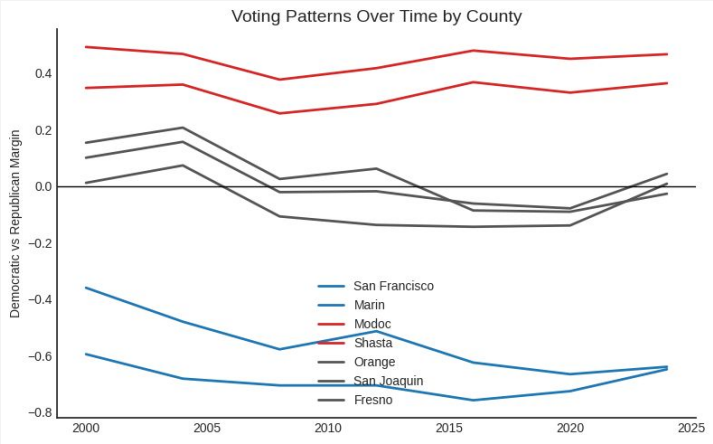
County-Level Voting Patterns



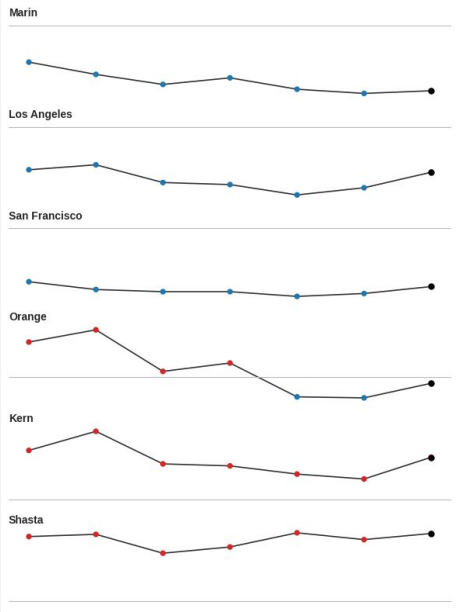
- Voting patterns vary significantly across counties
- Coastal regions show strong Democratic margins
- Inland regions consistently lean Republican
- Geographic differences challenge the “solid blue state” narrative

County-Level Voting Patterns

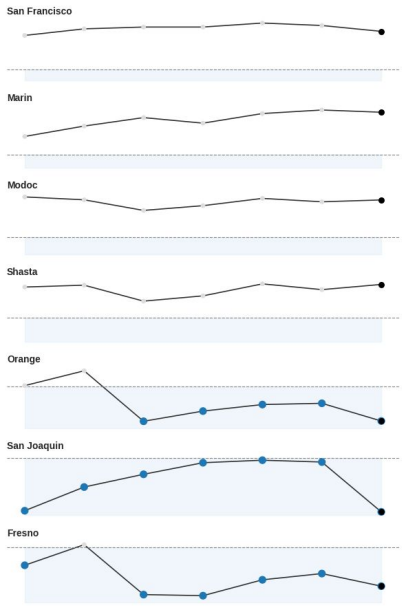
Voting Patterns



Margin

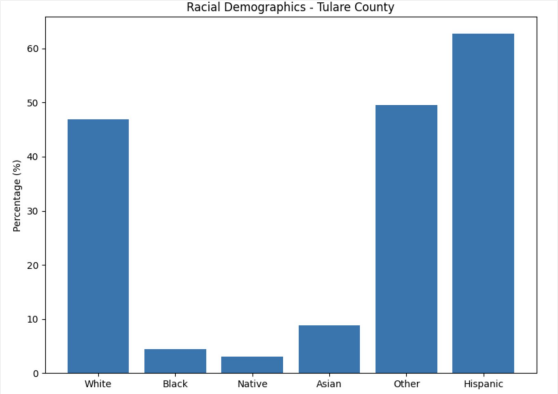


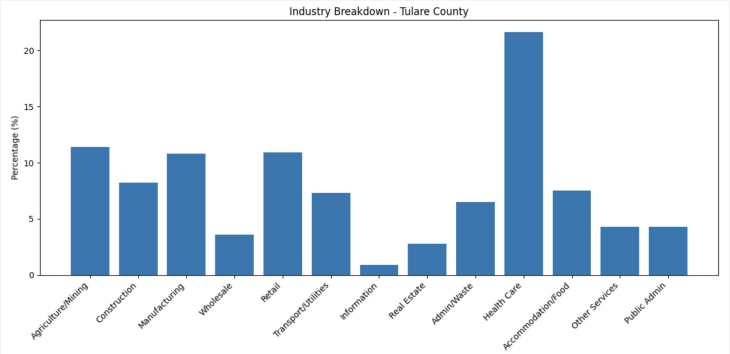
Competition



Geography does not explain everything, there are ___ and ___ factors

Tulare County





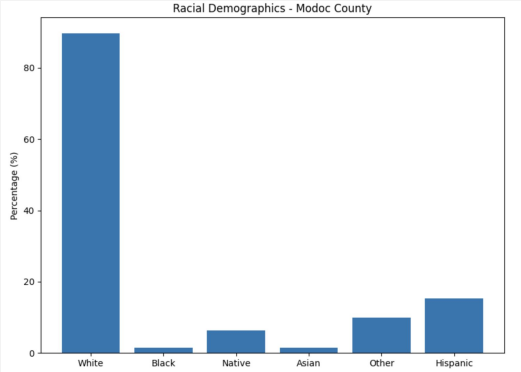
32

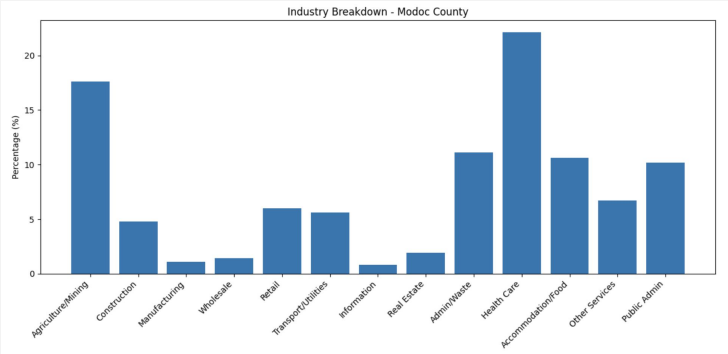
Median age

$64,474

Median household income

Modac County





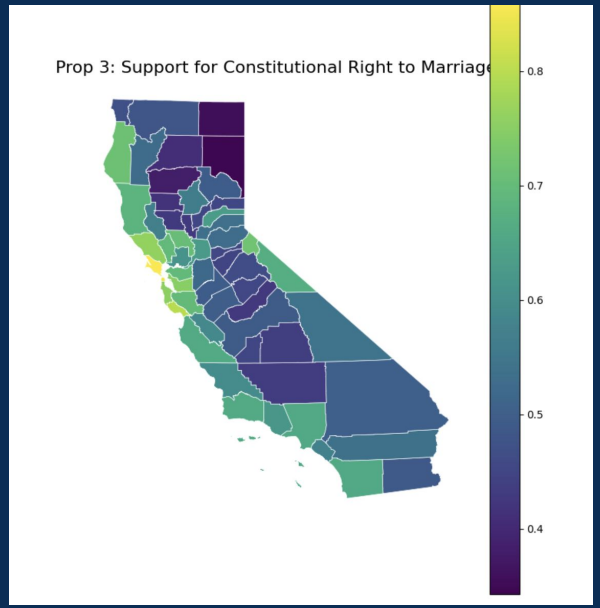
49

Median age

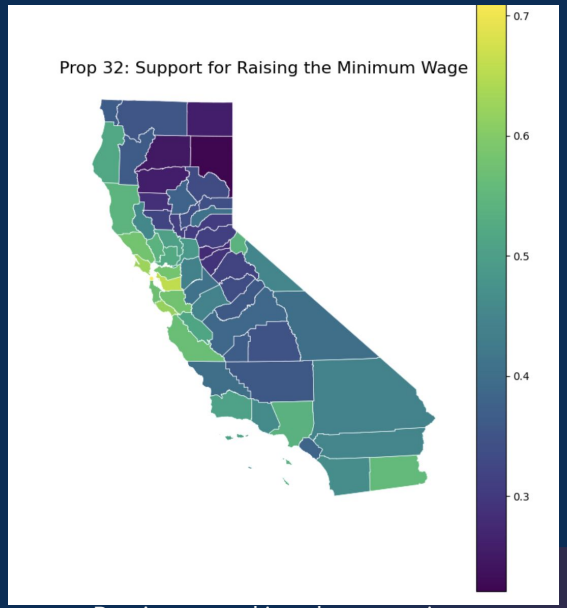
$54,962

Median household income

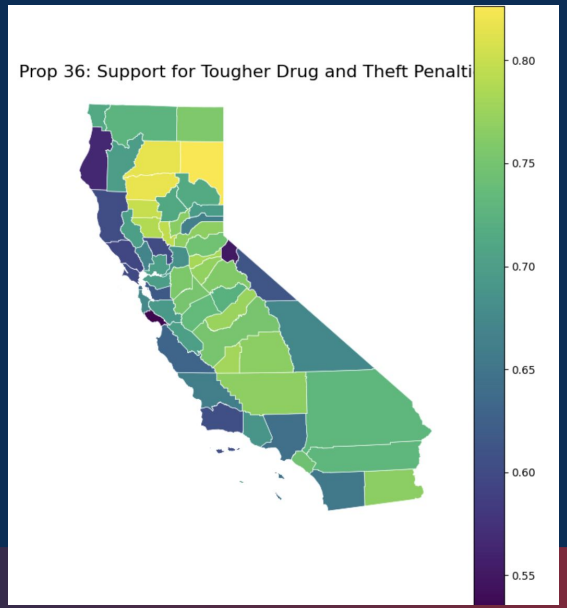
Counties Vote Differently Depending on the Issue



Bay Area strongest support



Bay Area + working-class counties



Rural/inland strongest support

County politics are multidimensional: social, economic, and criminal justice issues create different coalitions.

Why Proposition Voting Matters

What Are Propositions	Why Use Them	Focus Measures
California propositions are statewide ballot measures that allow voters to decide policy questions directly. Unlike candidate elections, propositions show how counties vote on specific issues rather than political parties alone.	Because every county votes on the same issue, propositions let us compare political preferences across California without the noise of candidate personality, incumbency, or campaign dynamics. They show what counties actually support when voting on policy itself.	**Prop 3:**Measured support for expanding civil rights and protecting same-sex marriage in the state constitution. **Prop 32 :** Measured support for government intervention to increase wages and strengthen protections for lower-income workers. **Prop 36:** Measured support for stricter law enforcement, harsher sentencing, and a more punitive approach to crime.

Findings and Next Steps

Key Findings and Next Steps

Findings So Far:

- California is blue statewide but politically mixed locally
- Geography matters, but not fully
- Counties vote differently by issue area
- Economic + demographic variables likely explain additional variation

Next Steps:

- Continue constructing county profiles
- Zoom out beyond individuals to explore which variables are most strongly associated with Republican vote share
- Build final polished visualizations